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Distributed economic dispatch for microgrids with vehicle-to-grid under time-varying network

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ABSTRACT

As distributed energy sources are increasingly integrated into power grids, centralized optimization approaches face challenges including limited robustness, high communication and computational overhead, and privacy concerns. Distributed economic dispatch algorithms, leveraging edge computing for local optimization and inter-agent communication for collaborative decision-making, offer a promising solution. However, existing methods often struggle to solve large-scale mixed-integer quadratic programming (MIQP) problems, time-coupled constraints, and plug-and-play optimization in vehicle-to-grid (V2G) systems. To address these challenges, this study simplifies the V2G system, modeled as an MIQP problem, into a convex quadratic optimization problem, and combines the alternating direction method of multipliers (ADMM) with model predictive control (MPC) to enable distributed solutions and plug-and-play capabilities. Lagrange multipliers are extended to a vector form to manage time-state coupling constraints. Physical constraints, including the constant current-constant voltage (CC-CV) charging mode and charge–discharge efficiency factors, are incorporated to enhance battery lifespan. A dispatch packet strategy reduces communication and computational overhead. Simulation and physical experiments demonstrate the proposed approach's feasibility and effectiveness.

1. Introduction

In microgrid operation, meeting energy demands requires determining the optimal generation schedule for available resources, which have diverse operational and cost characteristics. Achieving optimal microgrid performance involves solving an optimization problem encompassing economic and/or technical objectives. This problem incorporates soft constraints related to generation specifications and hard constraints ensuring the balance between production and demand. This process is known as microgrid economic dispatch [1,2].

In a microgrid, a central agent collects information from all system participants and determines their optimal actions based on an objective function. Centralized approaches typically aim to maximize global benefits, leveraging access to complete information. For example, recent AI-based heuristics, such as the African Vulture Optimizer [3,4], have demonstrated success in rapidly solving such centralized optimization problems. A centralized energy management approach for renewable-based distribution systems is introduced in [5]. Based on [5], an online reinforcement learning-based energy management strategy for microgrids is proposed in [6]. Considering the uncertainty of V2G owners' charging/discharging behavior and range anxiety, a V2G scheduling strategy suitable for the scale of charging stations was proposed in [7]. While such methods excel in making precise and comprehensive decisions, their stringent communication demands, central computational requirements, and privacy concerns can impede microgrid operations, especially in scenarios with numerous renewable energy sources and electric vehicle (EV) participants [8,9]. In contrast, distributed optimization scheduling architectures exchange only local decision variables with neighboring agents, resulting in reduced communication requirements and eliminating the need for a central control agent, such as a microgrid main controller, or associated communication links. However, the presence of coupling constraints and dynamic characteristics, which interconnect the decisions of individual controllers, makes the design of distributed optimization algorithms highly challenging [10–12].

While centralized optimization provides a global optimum, its practical application in large-scale microgrids faces significant hurdles. These include a high computational burden on the central agent, stringent communication requirements, and a lack of data privacy for participants [13]. To overcome these limitations, distributed optimization scheduling architectures have emerged as a necessary alternative. In this paradigm, agents exchange only local information with neighbors, which fundamentally enhances scalability, protects user privacy,

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and improves robustness against single points of failure. This shift in architecture, however, introduces new challenges in algorithm design, leading to various distributed convex optimization approaches. Depending on the nature of the distributed convex optimization problems, existing research can be primarily categorized into two types: differentiable convex optimization methods and non-differentiable convex optimization methods. In differentiable convex optimization methods, the Karush-Kuhn–Tucker (KKT) conditions are commonly used to decouple subproblems [14]. The main solutions include distributed gradient algorithms and consensus algorithms. [15] proposes a fully distributed consensus algorithm under weak assumptions, combining the convergence of incremental cost values with an innovative term that ensures power balance. [16] introduces an energy management algorithm for islanded microgrids, based on a fixed step-size improvement to existing consensus algorithms, which accelerates the convergence speed. [17] proposes a Push-based consensus method to solve a class of economic dispatch problems considering transmission line losses, which reduces the strong connectivity requirement of communication topologies, and enhances robustness under communication delays and constraint noise. Additionally, security [18] and privacy [19] concerns in distributed energy management based on consensus algorithms have received further attention. The implementation process is straightforward and was thoroughly explored in early theoretical studies. However, due to its limited design flexibility, it has not been widely adopted in practical applications of distributed energy management algorithms. In non-differentiable convex optimization methods, Lagrangian relaxation is commonly used to decouple the subproblems of non-differentiable convex optimization. Since traditional gradient methods are not applicable, subgradients are introduced to address energy management problems with non-differentiable cost functions [20]. Key solutions include projection algorithms [21], dual decomposition algorithms [22], and the alternating direction method of multipliers (ADMM) [13,23]. These methods are simple in terms of protocol and exhibit strong performance in both convergence speed and accuracy. Among them, ADMM combines the robustness of augmented Lagrange multipliers with the convergence benefits of dual decomposition and has gained significant attention in the optimization of smart grid operations [24].

A distributed algorithm leveraging ADMM for managing residential demand response programs is developed in [25]. In [26], a two-stage planning approach is proposed for coordinating integrated energy microgrids, combining game theory with ADMM. A robust, data-driven energy management framework for microgrids, using ADMM to improve scalability and privacy, is introduced in [27]. While these studies [26–28] address algorithm convergence and optimality, they assume static and reliable communication topologies. In reality, however, communication networks often experience disruptions, such as agent outages or link failures. To tackle these issues, [29] models the communication network as a random, time-varying undirected graph and presents a distributed algorithm for economic dispatch with transmission losses. To handle packet losses and delays in ADMM-based algorithms, virtual buffers and nodes are introduced in [30]. Similarly, a consensus-based method with virtual buffers for resolving communication link problems is discussed in [31]. Nonetheless, adding virtual buffers or nodes increases algorithmic complexity. Moreover, these approaches often overlook the diversity of distributed energy resources and the temporal dependencies across multiple time periods, which are essential for effectively managing vehicle-to-grid (V2G) systems. This is due to the limited storage capacity of EVs and the inherent temporal correlations in their state of energy (SOE). [32] considers the temporal correlations of SOE and solves a comprehensive energy sharing problem using the dual-consensus version of the ADMM algorithm, aiming to maximize social welfare through peer-to-peer energy cooperation. Building on the work of [32,33] proposes a fully distributed algorithm based on ADMM combined with model predictive control (MPC), meeting the plug-and-play requirements of EVs and considering battery charging efficiency. However, it does not account for V2G scenarios, and theoretical insights into the convergence and optimality of such distributed algorithms remain limited. Furthermore, these algorithms overlook the commonly used constant current-constant voltage (CC-CV) charging mode for lithium batteries. Consequently, during the later stages of charging, the battery voltage may exceed its rated limit, accelerating the aging process [7].

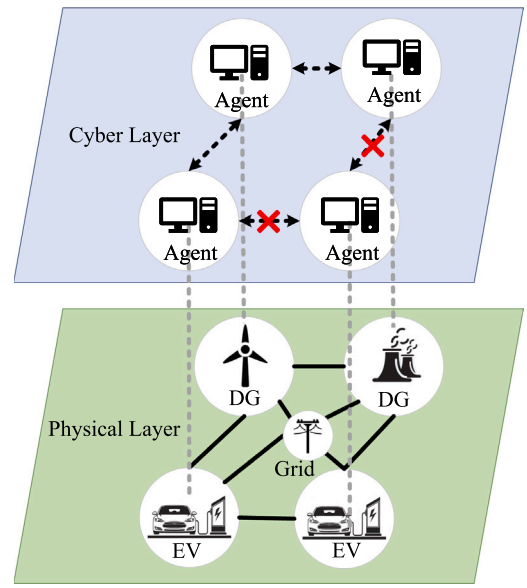

Fig. 1. Cyber–physical framework for distributed microgrids operations.

To address the aforementioned gaps in existing research, this paper proposes a fully distributed economic dispatch method that provides a robust, scalable, and computationally efficient solution for the real-time coordination of V2G systems based on ADMM and predictive control. The key contributions of this paper are summarized as follows:

1. The V2G MIQP problem is simplified into a convex quadratic optimization problem, solved by a distributed ADMM algorithm. An MPC-based dispatch packet strategy is introduced to handle plug-and-play EV arrivals and time-varying network requirements.
2. The algorithm incorporates the CC-CV charging mode and battery charging/discharging efficiency factors as key constraints to enhance battery longevity and address degradation concerns.
3. The algorithm's feasibility and effectiveness are validated through both extensive simulation and a physical hardware experimental prototype built with a distributed optimization network and battery charge/discharge system.

The notation used in this paper is standard. For a matrix A, its transpose is represented by A', and $\|A\|$ denotes the induced norm. The Kronecker product of matrices A and B is written as $A \otimes B$. The identity matrix of dimension $n \times n$ is denoted by I_n.

2. Preliminaries and problem formulation

This section presents the communication architecture and system model under consideration, along with the optimization challenges associated with time-varying network topologies, as shown in Fig. 1.

2.1. Time-Varying communication network

A *weighted undirected graph* $\mathcal{G}(t)$ is represented as a triple $(\Psi, \mathcal{E}(t), \mathcal{A}(t))$, where $\Psi = \{1, \dots, n\}$ is the set of agents, $\mathcal{E}(t)$ is a set of edges representing connections between agents, and $\mathcal{A}(t) = [\omega_{ij}] \in \mathbb{R}^{n \times n}$

is the adjacency matrix. Here, $\omega_{ij} > 0$ if $(i,j) \in \mathcal{E}(t)$, and $\omega_{ii} = 0$. If $(i,j) \in \mathcal{E}(t)$ implies $(j,i) \in \mathcal{E}(t)$ and $\omega_{ij} = \omega_{ji}$, the graph is *undirected*. An *undirected graph* $\mathcal{G}(t)$ is called *connected* if any two distinct nodes are linked by a path. The *Laplacian matrix* $L = [\ell_{ij}]$ for graph $\mathcal{G}(t)$ is defined by $\ell_{ii} = \sum_{j=1}^{n} \omega_{ij}$ and $\ell_{ij} = -\omega_{ij}$ for $i \neq j$.

Consider a time-varying network where agents may randomly switch between active and inactive states, and communication links are potentially unreliable. At each iteration, an agent $i \in \Psi$ is active with probability $\alpha_i \in (0,1]$, while each link $(i,j) \in \mathcal{E}(t)$ experiences a communication absence with probability $\beta_{ij} \in [0,1)$. Consequently, the probability of successful communication between agents i and j is given by $\gamma_{ij} = \alpha_i \alpha_j (1 - \beta_{ij})$. A link (i,j) is considered active if this condition is met.

Assumption 1. For a fixed t, the undirected graph $\mathcal{G}(t)$ is connected.

Assumption 1, a standard condition for consensus optimization [13], ensures that information from each agent persistently influences all other agents over time.

2.2. Problem of interest

Consider a microgrid network with $N + M$ agents, consisting of N distributed generators (DGs) and M EVs. The objective is to minimize total production costs while ensuring that the combined output of DGs and EVs meets the grid and charging demand over a finite discrete scheduling horizon $\mathcal{T} = \{1, \dots, T\}$. EVs arrive randomly, each with specific energy demands and deadlines. Given the limited storage capacity of EVs, coordinating their energy use across time steps is crucial. Efficient vehicle-to-grid utilization requires considering intertemporal constraints, addressing the uncertainty of EV arrivals, and optimizing the system over the time horizon to accommodate EV charging needs.

The total production cost, defined as the sum of microgrid costs over the time horizon, is given by:

$$\sum_{t=1}^{T} \sum_{i=1}^{N+M} C_i(P_i^t), \tag{1}$$

where P_i^t denotes the power generated by agent i at time t, and the cost $C_i(P_i^t)$ for each distributed generator or electric vehicle follows a quadratic function:

$$C_i(P_i^t) = a_i (P_i^t)^2 + b_i P_i^t + c_i, \tag{2}$$

with $a_i > 0$, $b_i > 0$, and $c_i > 0$ as the cost coefficients.

The power balance at each time step, requiring that the combined output of DGs and EVs matches the total demand, is expressed as:

$$\sum_{i=1}^{N+M} P_i^t = \sum_{i=1}^{N+M} D_i^t = D_{tot}^t \quad \forall t \in \mathcal{T}, \tag{3}$$

where D_i^t denotes the local demand of agent i, and D_{tot}^t represents the total demand at time t.

2.3. Modeling of distributed generators

For each DG $i \in \mathcal{N} := \{1, \dots, N\}$, the power output and ramping rate are subject to the following constraints:

$$P_i^{\min} \leq P_i^t \leq P_i^{\max} \quad \forall t \in \mathcal{T}, \forall i \in \mathcal{N}, \tag{4}$$

$$\Delta \underline{P_i} \leq P_i^t - P_i^{t-1} \leq \Delta \overline{P_i} \quad \forall t \in \mathcal{T}, \forall i \in \mathcal{N}, \tag{5}$$

where $P_i^{\min}$ and $P_i^{\max}$ denote the minimum and maximum power limits, while $\Delta \underline{P_i}$ and $\Delta \overline{P_i}$ represent the lower and upper ramping rate bounds, respectively.

To ensure system reliability, it is assumed that the total demand at each time step is fully met by DGs, leading to the following constraint:

$$\sum_{i=1}^{N} P_i^{\min} \leq D_{tot}^t \leq \sum_{i=1}^{N} P_i^{\max} \quad \forall t \in \mathcal{T}. \tag{6}$$

2.4. Modeling of electric vehicles

EVs remain idle for most of the day, providing significant flexibility for charging. Unlike traditional fixed-time charging, V2G technology facilitates bidirectional energy flow between EVs and the grid, creating a more intricate operational framework. The scheduling of EV charging and discharging, whether through standard energy transfers or V2G processes, is inherently time-dependent. This section introduces a V2G scheduling model that incorporates various constraints, including random arrival times, EV charging requirements, the CC-CV charging profile, maximum charging rates, energy transfer efficiencies, the mutual exclusivity of charging and discharging, and the distribution grid's aggregate charging power limits.

The key physical parameters for each EV $i \in \mathcal{M} := \{N+1, \dots, N+M\}$ are defined first. Each EV has a total energy capacity E_i^{cap} (in kWh) and a maximum power limit $P_i^{\max} > 0$ (in kW) for both charging and discharging. The energy conversion process is subject to discharging and charging efficiencies, η_i^{dis} and η_i^{cha}, respectively (where $0 < \eta \leq 1$). All operations are considered over discrete time steps of duration ΔT, corresponding to the horizon $\mathcal{T}$.

Given the operational constraint that electric vehicles cannot be charged and discharged simultaneously, the power flow at the grid connection point is defined as:

$$P_i^t = u_i^{t,\text{dis}} \cdot P_i^{t,\text{dis}} - u_i^{t,\text{cha}} \cdot P_i^{t,\text{cha}} \quad \forall t \in \mathcal{T}, \forall i \in \mathcal{M}, \tag{7}$$

where

$$0 \leq P_i^{t,\text{dis}}, P_i^{t,\text{cha}} \leq P_i^{\max} \quad \forall t \in \mathcal{T}, \forall i \in \mathcal{M}, \tag{8}$$

and

$$u_i^{t,\text{dis}} + u_i^{t,\text{cha}} \leq 1, \quad u_i^{t,\text{dis}}, u_i^{t,\text{cha}} \in \{0,1\} \quad \forall t \in \mathcal{T}, \forall i \in \mathcal{M}. \tag{9}$$

Here $P_i^{t,\text{dis}}$ and $P_i^{t,\text{cha}}$ are the discharging and charging power variables, while $u_i^{t,\text{dis}}$ and $u_i^{t,\text{cha}}$ are binary decision variables indicating discharge and charge activation states.

The internal battery power, which accounts for the previously defined efficiency losses, dictates that storing 1 kWh of energy during off-peak hours yields $\eta_i^{dis}\eta_i^{cha}$ kWh of usable energy during peak hours. This is modeled as:

$$P_i^{t,\text{bat}} = \frac{1}{\eta_i^{dis}} P_i^{t,dis} - \eta_i^{cha} P_i^{t,cha}, \quad \forall t \in \mathcal{T}, \forall i \in \mathcal{M}, \tag{10}$$

where $P_i^{t,\text{bat}}$ is the net rate of energy change inside the battery of storage i at time t.

The state of energy (SOE) for EV i at time t evolves based on the internal battery power rate, the time step duration ΔT, and the total capacity E_i^{cap} as follows:

$$SOE_i^t = SOE_i^{t-1} - \frac{P_i^{t,\text{bat}} \Delta T}{E_i^{\text{cap}}} = \frac{E_i^t}{E_i^{\text{cap}}}, \quad \forall t \in \mathcal{T}, \ \forall i \in \mathcal{M}, \tag{11}$$

where E_i^t denotes the total energy (in kWh) stored in EV i at time t.

At the start of interval t_o, arriving EVs with initial SOE ($SOE_i^{t_o}$) at charger i reveal stochastic required departure time SOE (SOE_i^{dep}) and departure time (T_i^{dep}). When the departure time is reached, EVs are removed from the system. The SOE must meet the required level upon departure:

$$\begin{cases} SOE_i^t = SOE_i^{t_o} & \text{if } t = t_o \\ 0 \leq SOE_i^t \leq 1 & \text{if } t_o \leq t \leq T_i^{dep} \\ SOE_i^t \geq SOE_i^{dep} & \text{if } t \geq T_i^{dep}. \end{cases} \tag{12}$$

To ensure battery longevity, the CC-CV charging mode for lithium-ion batteries is employed. The charging begins with a constant current (CC) phase, during which the cell voltage rises. During the CC phase, a constant maximum charging power $P_i^{\max}$ is applied. Once the voltage reaches the upper limit, $V_{\max}$, at a specific SOE (SOE_i^{swi}), the mode

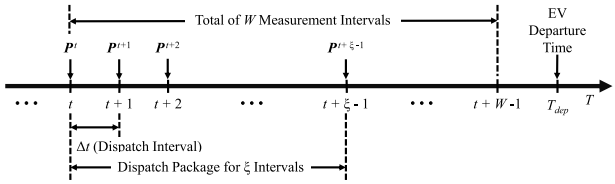

Fig. 2. Temporal dispatch packet strategy structure of the proposed MPC algorithm.

switches to constant voltage (CV). In the CV phase, the voltage remains steady while the current decreases. This behavior is approximated using a piecewise linear model based on empirical data [34]. The maximum charging power profile is constrained by the SOE as follows:

$$P_i^{t,\max} = \begin{cases} P_i^{\max}, & \text{if } SOE_i^t \le SOE_i^{swi}, \\ \frac{P_i^{\max}\cdot(1-SOE_i^t)}{1-SOE_i^{swi}}, & \text{if } SOE_i^t > SOE_i^{swi}, \end{cases} \tag{13}$$

where $P_i^{t,\max}$ represents the time-varying upper limit for charging power of EV i at time t, and SOE_i^{swi} is the transition point. This constraint must be respected by the charging variable, such that $P_i^{t,\text{cha}} \le P_i^{t,\max}$ for all $t \in \mathcal{T}$. In this study, SOE_i^{swi} is set to 85%, as noted in [7].

3. Proposed distributed economic dispatch approach

To mitigate the impact of the unpredictability of charging behavior during V2G operations for EVs, an MPC rolling optimization framework is adopted. When a vehicle arrives randomly, the system triggers a new optimization cycle to incorporate the newly arrived vehicle. Similarly, when a vehicle unexpectedly leaves, the optimization process is updated to exclude the departed vehicle. This iterative process helps to reduce the impact of charging behavior uncertainty on V2G scheduling.

MPC features a rolling optimization procedure with a look-ahead horizon. Traditional MPC methods compute the optimal dispatch sequence for the entire horizon at each time step but implement only the first element, which can be computationally expensive. To address this, inspired by the work in [35–37], a dispatch packet strategy is proposed. This strategy, common in networked control systems, helps mitigate issues like delays and packet loss. By applying this approach to the distributed economic dispatch algorithm, both computational and communication loads are reduced. As shown in Fig. 2, during the MPC rolling optimization process, each computation packet contains optimized scheduling signals for the time interval from t to $t+\xi-1$, where $\xi \le W$ (with ξ representing the control packet length and W the prediction time horizon). Re-optimization occurs only when a new vehicle arrives or at time $t+\xi-1$. By combining the scheduling packet strategy with rolling optimization, the impact of random charging demand arrivals is minimized while computational and communication demands are alleviated.

Given the proposed model and the limited capacity of energy storage in EVs, the operation of V2G over time is interdependent. Therefore, optimizing their use across multiple time steps is essential. To address these intertemporal constraints and improve the efficiency of EV operations, the following multistep economic dispatch problem is formulated:

$$\min_{P_i^t, P_i^{t,dis}, P_i^{t,cha}, SOE_i^t} J(t) = \sum_{h=t}^{t+W-1}\sum_{i=1}^{N+M} C_i(P_i^h) \tag{14}$$
$$\text{s.t.} \quad (3)\ (4)\ (5)\ (6)\ (7)\ (8)\ (9)\ (10)\ (11)\ (13)$$

where W refers to the predicted duration of time. Constraints (7) and (9) transform the optimization problem defined by (14) into an MIQP formulation. Additionally, the inclusion of charging and discharging efficiency coefficients in (10) prevents direct simplification of the model. However, constraint (7) can be equivalently expressed as $P_i^{t,\text{dis}} \cdot P_i^{t,\text{cha}} = 0$, which enforces that at least one of $P_i^{t,\text{cha}}$ or $P_i^{t,\text{dis}}$ must be zero during each time period, thereby converting the integer constraint into a nonlinear formulation.

Remark 1. To prevent simultaneous charging and discharging, either $P_i^{t,dis}$ or $P_i^{t,cha}$ must be zero at any given time:

$$P_i^{t,dis} \cdot P_i^{t,cha} = 0 \quad \forall t \in \mathcal{T}, \forall i \in \mathcal{M}. \tag{15}$$

However, when η_i^{dis} and η_i^{cha} are strictly less than 1, this nonconvex constraint (15) can be omitted from the optimization problem [38]. This is because any optimal solution will inherently satisfy this condition to avoid efficiency losses, as formally proven in Appendix.

When (15) is omitted, constraints (7) and (9) reduce to the linear form $P_i = P_i^{\text{dis}} - P_i^{\text{cha}}$. Consequently, (14) simplifies to a convex quadratic optimization problem, eliminating the duality gap while ensuring non-empty dual optimal solution sets.

The Lagrange dual problem of (14) is formulated as

$$\min_{\lambda \in \mathbb{R}^W} \sum_{i=1}^{N+M} \Phi_i(\lambda) \tag{16}$$

where

$$\Phi_i(\lambda) = \max_{P_i^t \in \Omega_i} \sum_{h=t}^{T+W-1} -\left[C_i(P_i^h) + \lambda^h(P_i^h - D_i^h)\right] \quad i \in \Psi, \tag{17}$$

and $\lambda \in \mathbb{R}^W$ is the Lagrange multiplier associated with the power balance constraints (3), with $\lambda = (\lambda^t, \dots, \lambda^{t+W-1})'$. To enable distributed optimization, each agent i is assigned a local copy of λ, denoted by λ_i. A consensus constraint ensures the uniformity of λ_i values across agents. Consequently, problem (16) can be reformulated as

$$\begin{aligned} \min_{\lambda_i \in \mathbb{R}^\prime} \quad & \sum_{i=1}^{N+M} \Phi_i(\lambda_i) \\ \text{s.t.} \quad & \lambda_i = z_{ij},\ \lambda_j = z_{ij} \quad \forall j \in \mathcal{N}_i, \quad i \in \Psi, \end{aligned} \tag{18}$$

where z_{ij} are the consensus auxiliary variables. The constraints in (18) enforce neighbor-wise consensus, ensuring that $\lambda_i = \lambda_j$ for all $j \in \mathcal{N}_i$ and $i \in \Psi$. Under Assumption 1, achieving consensus between neighbors results in global consensus, making problem (18) equivalent to problem (16).

By applying the multi-agent distributed ADMM algorithm [33] to problem (18), the augmented Lagrangian for (18) is expressed as

$$\begin{aligned} \mathcal{L} = & \sum_{i=1}^{N+M} \Phi_i(\lambda_i) + \sum_{i=1}^{N+M}\sum_{j\in\mathcal{N}_i} \left(\mu_{ij}'(\lambda_i - z_{ij}) + \nu_{ij}'(\lambda_j - z_{ij})\right) \\ & + \frac{d}{2}\sum_{i=1}^{N+M}\sum_{j\in\mathcal{N}_i}\left(\left\|\lambda_i - z_{ij}\right\|_2^2 + \left\|\lambda_j - z_{ij}\right\|_2^2\right), \end{aligned} \tag{19}$$

where $\mu_{ij} \in \mathbb{R}^W$ and $\nu_{ij} \in \mathbb{R}^W$ are Lagrange dual variables associated with the constraints in (18), and $d > 0$ is the penalty parameter. The ADMM updates for solving (19) are iterated for $k = 0, 1, \dots$, with each update performed as follows:

$$\mu_{ij}^{(k+1)} = \mu_{ij}^{(k)} + \frac{d}{2}\left(\lambda_i^{(k)} - \lambda_j^{(k)}\right) \quad \forall j \in \mathcal{N}_i,\ i \in \Psi, \tag{20a}$$

$$\nu_{ij}^{(k+1)} = \nu_{ij}^{(k)} + \frac{d}{2}\left(\lambda_j^{(k)} - \lambda_i^{(k)}\right) \quad \forall j \in \mathcal{N}_i,\ i \in \Psi, \tag{20b}$$

$$\begin{aligned} \lambda_i^{(k+1)} = & \arg\min_{\lambda_i} \left\{ \Phi_i(\lambda_i) + \lambda_i'\left(\mu_{ij}^{(k+1)} + \nu_{ji}^{(k+1)}\right) \right. \\ & \left. + d\sum_{j\in\mathcal{N}_i}\left\|\lambda_i - \frac{\lambda_i^{(k)} + \lambda_j^{(k)}}{2}\right\|_2^2 \right\} \\ = & \arg\min_{\lambda_i} \max_{P_i \in \Omega_i} \left\{ \sum_{h=t}^{t+W-1} -[C_i(P_i^h) + \lambda^h(P_i^h - D_i^h)] \right. \end{aligned}$$

$$+\lambda_i'(\mu_{ij}^{(k+1)}+\nu_{ji}^{(k+1)})+d\sum_{j\in\mathcal{N}_i}\left\|\lambda_i-\frac{\lambda_i^{(k)}+\lambda_j^{(k)}}{2}\right\|_2^2\Bigg\}. \tag{20c}$$

Solving subproblem (20c) is challenging due to its min–max optimization structure. However, since the objective function in (14) is convex in one variable and concave in the other, the minimax theorem [39] can be applied. This allows the min–max problem (20c) to be equivalently transformed into its max–min counterpart, ensuring the existence of a saddle point. The max–min formulation of (20c) is expressed as:

$$\begin{aligned}
&\max_{P_i\in\Omega_i}\min_{\lambda_i}\Bigg\{\sum_{h=t}^{t+W-1}-[C_i(P_i^h)+\lambda^h(P_i^h-D_i^h)]\\
&\quad+\lambda'(\mu_{ij}^{(k+1)}-\nu_{ji}^{(k+1)})+d\sum_{j\in\mathcal{N}_i}\left\|\lambda_i-\frac{\lambda_i^{(k)}+\lambda_j^{(k)}}{2}\right\|_2^2\Bigg\}\\
&=\max_{P_i\in\Omega_i}\min_{\lambda_i}\Bigg\{\sum_{h=t}^{t+W-1}\Bigg[-C_i(P_i^h)-\frac{d}{4|\mathcal{N}_i|}\Bigg\|\frac{1}{d}(P_i^h-D_i^h)\\
&\quad-\frac{1}{d}(\mu_{ij}^{h,(k+1)}+\nu_{ji}^{h,(k+1)})+\sum_{j\in\mathcal{N}_i}(\lambda_i^{h,(k)}+\lambda_j^{h,(k)})\Bigg\|_2^2\Bigg]\\
&\quad+(d|\mathcal{N}_i|)\Bigg\|\lambda_i-\frac{1}{2|\mathcal{N}_i|}\Bigg[\sum_{j\in\mathcal{N}_i}(\lambda_i^{(k)}+\lambda_j^{(k)})\\
&\quad+\frac{1}{d}(P_i^{(k+1)}-D_i)-\frac{1}{d}(\mu_{ij}^{(k+1)}+\nu_{ji}^{(k+1)})\Bigg]\Bigg\|_2^2\Bigg\}.
\end{aligned} \tag{21}$$

The equality is attained by completing the square of the quadratic term. Let $P_i^{(k+1)}$ denote the inner maximizer of (20c), such that $(P_i^{(k+1)},\lambda_i^{(k+1)})$ forms a saddle point of (20c). Thus, $\lambda_i^{(k+1)}$ acts as the outer–inner solution to (20c).

Following [40], by initializing $\mu_{ij}^{(0)}+\upsilon_{ji}^{(0)}=0\quad\forall i,j\in\Psi$, the iterative formulas can be further simplified as

$$g_i^{(k+1)}=g_i^{(k+1)}+d\sum_{j\in\mathcal{N}_i}\left(\lambda_i^{(k)}-\lambda_j^{(k)}\right) \tag{22}$$

$$\begin{aligned}
P_i^{(k+1)}=&\arg\min_{P_i\in\Omega_i}\Bigg\{\sum_{h=t}^{t+W-1}\Bigg[C_i(P_i^h)+\frac{d}{4|\mathcal{N}_i|}\Bigg\|\frac{1}{d}(P_i^h-D_i^h)\\
&-\frac{1}{d}g_i^{h,(k+1)}+\sum_{j\in\mathcal{N}_i}(\lambda_i^{h,(k)}+\lambda_j^{h,(k)})\Bigg\|_2^2\Bigg]\Bigg\}
\end{aligned} \tag{23}$$

$$\begin{aligned}
\lambda_i^{(k+1)}=&\frac{1}{2|\mathcal{N}_i|}\Bigg[\sum_{j\in\mathcal{N}_i}(\lambda_i^{(k)}+\lambda_j^{(k)})-\frac{1}{d}g_i^{(k+1)}\\
&+\frac{1}{d}(P_i^{(k+1)}-D_i)\Bigg]
\end{aligned} \tag{24}$$

where, $g_i^{(k+1)}=\sum_{j\in\mathcal{N}_i}\mu_{ij}^{(k)}+\upsilon_{ji}^{(k)}$. As a result, the proposed distributed economic dispatch algorithm can be decomposed across agents and updated in parallel.

Distributed optimization in practical applications involves a dynamic network, where agents may randomly transition between active and inactive states, and communication links may be unreliable. Thus, a distributed economic dispatch algorithm that can withstand such dynamic network conditions is essential. The proposed algorithm adopts a robust approach as outlined in [41]. At iteration k, let $\Omega^k\subseteq\Psi$ denote the set of active agents, and let $\Phi^k\subseteq\{(i,j)\in\mathcal{E}(t)\mid i,j\in\Omega^k\}$ represent the set of active communication links. During each iteration k, only the active agents update their dual variables λ_i, and communication takes place solely between active agents connected by operational links. For each pair $(i,j)\in\mathcal{E}(t)$, let Ξ_{ij} and Y_{ij} represent the sum and difference of λ_i and λ_j, respectively. Their updates are given by:

$$\Xi_{ij}^{(k)}=\begin{cases}\lambda_i^{(k)}+\lambda_j^{(k)} & \text{if } (i,j)\in\Phi^k,\\ \lambda_i^{(k-1)}+\lambda_j^{(k-1)} & \text{if } \{(i,j)\notin\Phi^k\mid i,j\in\Omega^k\},\\ \text{Skip} & \text{if } i\notin\Omega^k\end{cases} \tag{25}$$

Algorithm 1 Proposed Distributed Economic Dispatch Algorithm with Time-Varying Network

Initialization Initialize system with ξ, W.
for $t\in\{0,1,2,\ldots,T\}$ **do**
  if t ≥ 1 **then**
    Use the previous state's optimization results as the initial value for the current optimization.
  end if
  if $\sigma>\xi$ **or** EV arrivals **then**
    Given load D_i^t and the charging demand of newly arrived EVs, $SOE_i^{t_o}$, SOE_i^{dep}, T_i^{dep}.
    Initialize variables $P_i^{(0)}\in\Omega_i$, $\lambda_i^{(0)}\in\mathbb{R}$, and $h_i^{(0)}=0$ for each agent $i\in\Psi$. Set $k=1$.
    while not converged **do**
      for $i\in\Omega^k$ (in parallel) **do**
        Compute Ξ_{ij} and Y_{ij} using (25) and (26), respectively;
        Update local variables g_i^{k+1}, P_i^{k+1} by solving (27) and (28);
        Update dual variables λ_i^{k+1} using (29);
        Broadcast λ_i^{k+1} to neighbors $j\in\mathcal{N}_i$ and receive λ_j^{k+1};
        if stopping criterion is satisfied **then**
          Declare convergence.
        end if
      end for
      for $i\notin\Omega^k$ (in parallel) **do**
        Update variables: $g_i^{k+1}=g_i^k$, $P_i^{k+1}=P_i^k$, $\lambda_i^{k+1}=\lambda_i^k$.
      end for
      Set $k=k+1$.
    end while
    Extract the first ξ intervals of economic dispatch from the solution (28);
    Refresh dispatch package counter σ: $\sigma=1$.
  end if
  Update dispatch package counter σ: $\sigma=\sigma+1$.
end for

$$Y_{ij}^{(k)}=\begin{cases}\lambda_i^{(k)}-\lambda_j^{(k)} & \text{if } (i,j)\in\Phi^k,\\ \lambda_i^{(k-1)}-\lambda_j^{(k-1)} & \text{if } \{(i,j)\notin\Phi^k\mid i,j\in\Omega^k\}.\\ \text{Skip} & \text{if } i\notin\Omega^k\end{cases} \tag{26}$$

Substituting Eqs. (25) (26) into Eqs. (22), (23) and (24) yields

$$g_i^{(k+1)}=g_i^{(k+1)}+d\sum_{j\in\mathcal{N}_i}Y_{ij}^{(k)}\quad\forall j\in\mathcal{N}_i,\ i\in\Omega^k \tag{27}$$

$$\begin{aligned}
P_i^{(k+1)}=&\arg\min_{P_i\in\Omega_i}\Bigg\{\sum_{h=k}^{t+W-1}\Bigg[C_i(P_i^h)+\frac{d}{4|\mathcal{N}_i|}\Bigg\|\frac{1}{d}(P_i^h-D_i^h)\\
&-\frac{1}{d}g_i^{t,(k+1)}+\sum_{j\in\mathcal{N}_i}\Xi_{ij}^{(k)}\ \|_2^2\Bigg]\Bigg\}
\end{aligned} \tag{28}$$

$$\lambda_i^{(k+1)}=\frac{1}{2|\mathcal{N}_i|}\Bigg[\sum_{j\in\mathcal{N}_i}\Xi_{ij}^{(k)}-\frac{1}{d}g_i^{(k+1)}+\frac{1}{d}(P_i^{(k+1)}-D_i)\Bigg]. \tag{29}$$

The proposed methodology is detailed in Algorithm 1. It integrates a MPC rolling-horizon framework with a distributed ADMM core to create a robust and scalable solution. This combined approach is specifically designed to address the key challenges of V2G dispatch: it uses MPC to manage the uncertainty of plug-and-play EV arrivals, while employing a distributed ADMM to mitigate the computational load and privacy concerns inherent in centralized methods. The following details the precise workflow, data flow, and synchronization mechanisms that

combine these two strategies. The synergy between MPC and ADMM is realized through a structured rolling optimization cycle. The control flow begins at each MPC time step, where agents perform a distributed ADMM optimization based on the current system state and predicted load/arrival data for the upcoming horizon. This constitutes the primary data flow into the optimization. Once the ADMM converges, only the initial segment of the optimal schedule, specifically the dispatch packet of length ξ, is executed, and control commands are issued. The system then advances in time. The synchronization mechanism triggers the next optimization cycle either after the dispatch packet's duration has elapsed or upon an external event, such as the arrival of a new EV. The resulting system state, for example, the updated EV SOE, serves as the feedback and initial condition for the subsequent ADMM optimization. This iterative process, where ADMM solves the optimization within each discrete time window and MPC rolls the horizon forward based on real-world outcomes, forms the core of the combined control and optimization strategy. The convergence analysis of the optimal distributed economic dispatch algorithm in time-varying networks is provided in Theorem 1. Once the convergence and optimality conditions are met in time-varying networks, these properties hold for time-invariant networks as well, as the latter can be considered a special case of the former. Therefore, this algorithm is more robust compared to those that consider only time-invariant networks.

Theorem 1. *Under Assumption 1, and with $\alpha_i > 0, \forall i \in \Psi$ and $\beta_{ij} < 1$, let $P_i^\star$ and $\lambda_i^\star$ denote the optimal primal and dual solutions of Problem (14). Then, the proposed distributed economic dispatch algorithm will converge to these optimal solutions.*

Proof. To prove Theorem 1, assume the values $h_i^k,\ P_i^k,\ \lambda_i^k$ at iteration k are known. For iteration $k+1$, the following updates are considered:

$$\tilde{g}_i^{(k+1)} = g_i^{(k+1)} + d \sum_{j\in\mathcal{N}_i} Y_{ij}^{(k)} \quad \forall j \in \mathcal{N}_i,\ i \in \Psi \tag{30}$$

$$\tilde{P}_i^{(k+1)} = \arg\min_{P_i\in\Omega_i} \left\{ \sum_{h=k}^{t+W-1} \left[C_i(P_i^h) + \frac{d}{4|\mathcal{N}_i|} \left\| \frac{1}{d}(P_i^h - D_i^h) - \frac{1}{d} g_i^{t,(k+1)} + \sum_{j\in\mathcal{N}_i} \Xi_{ij}^{(k)} \right\|_2^2 \right] \right\} \tag{31}$$

$$\tilde{\lambda}_i^{(k+1)} = \frac{1}{2|\mathcal{N}_i|} \left[\sum_{j\in\mathcal{N}_i} \Xi_{ij}^{(k)} - \frac{1}{d} g_i^{(k+1)} + \frac{1}{d}(P_i^{(k+1)} - D_i) \right]. \tag{32}$$

The complete iterations in Algorithm 1 are identical for all $i \in \Omega^k$ and $(i,j) \in \Phi^k$, i.e.,

$$\tilde{g}_i^{(k+1)} = g_i^{(k+1)},\ \tilde{P}_i^{(k+1)} = P_i^{(k+1)},\ \tilde{\lambda}_i^{(k+1)} = \lambda_i^{(k+1)}.$$

As the proposed distributed economic dispatch method leverages distributed consensus ADMM to address the dual problem in (18), it follows from [40] that, as $k \to \infty$,

$$\tilde{\lambda}_i^{(k+1)} \to \lambda^\star, \quad \tilde{\lambda}_i^{(k+1)} - \tilde{\lambda}_j^{(k+1)} \to 0, \quad \forall j \in \mathcal{N}_i,\ i \in \Psi. \tag{33}$$

It is sufficient to show that any limit point of $\{\tilde{P}_1^{(k+1)}, \tilde{P}_2^{(k+1)}, \ldots, \tilde{P}_{N+M}^{(k+1)}\}$ is asymptotically optimal for problem (18), i.e., as $k \to \infty$,

$$\partial C_i(\tilde{P}_i^{(k+1)}) + \lambda^\star \to 0, \quad \forall i \in \Psi, \tag{34}$$

and fulfills the power balance constraints at each time step:

$$\sum_{i=1}^{N+M} \tilde{P}_i^{t,(k+1)} - \sum_{i=1}^{N+M} D_i^t \to 0. \tag{35}$$

In order to verify (34), consider the optimality condition of (23):

$$0 = \sum_{h=t}^{t+W-1} (\partial C_i(\tilde{P}_i^{h,(k+1)})) + \frac{1}{2|\mathcal{N}_i|}\left(\frac{1}{d}(\tilde{P}_i^{h,(k+1)} - D_i^h) - \frac{1}{d} g_i^{h,(k)} + \sum_{j\in\mathcal{N}_i} \Xi_{ij}^{(k)}\right). \tag{36}$$

By substituting (29) into (36), it follows that

$$0 = \partial C_i(\tilde{P}_i^{(k+1)}) + \tilde{\lambda}_i^{(k+1)}. \tag{37}$$

Since (37) holds for all k and $\tilde{\lambda}_i^{(k+1)} \to \lambda^\star$ as shown by (33), it follows that (37) remains true as $k \to \infty$.

To prove the power balance (34), rewrite (29) as follows:

$$\begin{aligned} 0 =& 2|\mathcal{N}_i| d\tilde{\lambda}_i^{(k+1)} - d \sum_{j\in\mathcal{N}_i} \Xi_{ij}^{(k)} + \tilde{h}_i^{(k+1)} - (\tilde{P}_i^{(k+1)} - D_i) \\ =& 2c \sum_{j\in\mathcal{N}_i} \left(\tilde{\lambda}_i^{(k+1)} - \frac{\Xi_{ij}^{(k+1)}}{2}\right) + \tilde{g}_i^{(k+1)} - (\tilde{P}_i^{(k+1)} - D_i) \\ &+ d \sum_{j\in\mathcal{N}_i} (Y_{ij}^{(k+1)} - Y_{ij}^{(k)}) \\ =& \tilde{h}_i^{(k+2)} - (\tilde{P}_i^{(k+1)} - D_i) + d \sum_{j\in\mathcal{N}_i} (Y_{ij}^{(k+1)} - Y_{ij}^{(k)}) \end{aligned} \tag{38}$$

By summing (38) over all DGs and EVs, and considering both the initialization and iterative update of h_i in (27), it follows that $\sum_{i=1}^{N+M} g_i^{(k)} = 0$. This leads to:

$$\sum_{i=1}^{N+M} (\tilde{P}_i^{(k+1)} - D_i) = d \sum_{i=1}^{N+M} \sum_{j\in\mathcal{N}_i} (Y_{ij}^{(k+1)} - \Delta_{ij}^{(k)}). \tag{39}$$

Since $\tilde{\lambda}_i^{(k+1)} - \tilde{\lambda}_i^{(k)} \to 0$ as $k \to \infty$, and $\tilde{\lambda}_i^{(k+1)} \to \lambda^*$ for all $i \in \Psi$, substituting this into (39) gives:

$$\sum_{i=1}^{N+M} (\tilde{P}_i^{(k+1)} - D_i) = 0. \tag{40}$$

This shows that the primal feasibility of $\tilde{P}_i^{(k+1)}$ for (16) depends on achieving consensus among the agents on the dual variable λ. By Theorem 1, the proposed distributed economic dispatch algorithm for time-varying networks guarantees optimal economic dispatch, thus completing the proof.

4. Case study

The purpose of this section is to verify the effectiveness and advantages of the proposed distributed economic dispatch algorithm. The designed algorithm supports the plug-and-play functionality of EVs and ensures convergence in time-varying networks. The IEEE 39-bus system, consisting of 4 DGs and 4 EVs, is employed to validate the proposed approach. It is assumed that there are 8 agents within the microgrid, with each agent corresponding to either a DG or an EV. The evolution of the communication topology of the microgrid is illustrated in Fig. 3. Both EV1 and EV2 can simultaneously plug into the microgrid for charging and V2G operations. After meeting the charging requirements of the EVs, they can be unplugged from the microgrid at any time. The parameters of the DGs and EVs are provided in Tables 1 and 2, respectively. The charging and discharging efficiencies are denoted as η_i^{cha} and η_i^{dis}, respectively. t_o and t_{dep} represent the arrival time and departure time of the EV, while $SOE_i^{t_o}$ and SOE_i^{dep} represent the state of energy at the time of arrival and the expected SOE at the time of departure, respectively. The storage capacities of EV1, EV2, and EV3 are 150 kWh, while the storage capacity of EV4 is 120 kWh. The penalty parameter d is set to 0.03, and the convergence error threshold is set to 10^{-5}.

The total power demand supplied without V2G is represented by the solid red line in Fig. 4. Algorithms 1 are used to coordinate 4 DGs and 4 EV units over a 24-h period. The resulting net load with V2G, defined as the load minus the EVs' output, is shown as the dashed blue line in the same figure. This demonstrates how the utilization of V2G effectively mitigates load fluctuations, contributing to peak shaving and valley filling in the power grid.

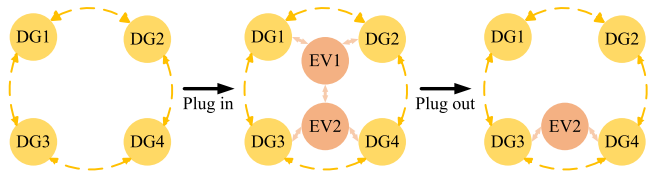

Fig. 3. The evolution of the communication topology when EV1 and EV2 are inserted and disconnected from the microgrid.

Table 1
Parameters of DGs [42].

DGi	a_i	b_i	c_i	P_i^{min}	P_i^{max}
1	0.120	3.3	25	10	60
2	0.124	3.3	28	13	140
3	0.140	4.5	27	14	200
4	0.118	3.8	26	19	250

Table 2
Parameters of EVs.

EVi	a_i	P_i^{min}	P_i^{max}	η_i^{cha}	η_i^{dis}	t_o	$SOE_i^{t_o}$	t_{dep}	SOE_i^{dep}
1	0.05	0	30	0.95	0.95	3	0.2	11	0.8
2	0.02	0	50	0.88	0.88	9	0.4	19	1
3	0.02	0	50	0.95	0.95	9	0.4	19	1
4	0.04	0	40	0.88	0.88	12	0.3	21	0.9

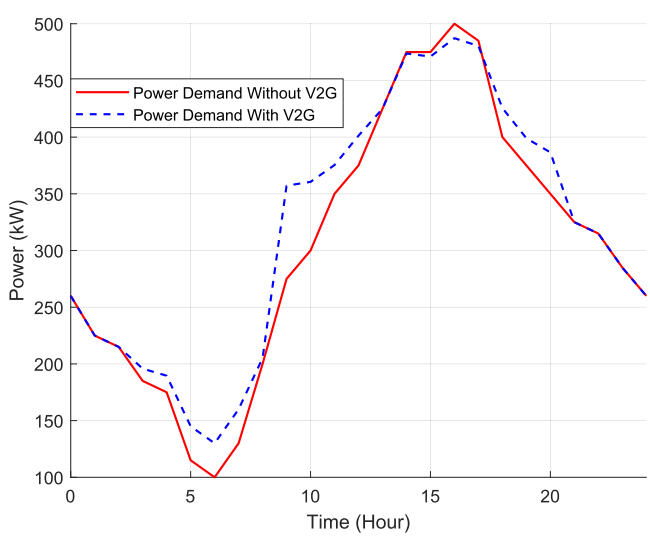

Fig. 4. The total power demand with or without V2G.

4.1. Time-varying network with compensation

In this case, the impact of a time-varying network is considered, with both the probabilities of agent disconnection and communication link failure set to 0.02. Compensation mechanisms are applied to address the dynamic nature of the network. Fig. 5(a) illustrates the optimal generation schedules for four distributed generators and four electric vehicles over a 24-h period. The coordinated operation of electric vehicles not only fulfills their charging requirements but also effectively mitigates peak loads and fills valleys by discharging during high-load periods and charging during low-load periods. EV2 and EV3 share identical parameters except for their charging and discharging efficiencies. Specifically, the charging efficiency (η_i^{cha}) and discharging efficiency (η_i^{dis}) are 0.88 for EV2 and 0.95 for EV3, indicating that EV3 has higher efficiency. Due to its greater efficiency, as shown in Fig. 5(a), EV3 discharges energy into the grid, whereas EV2 does not. This demonstrates that the proposed algorithm can design

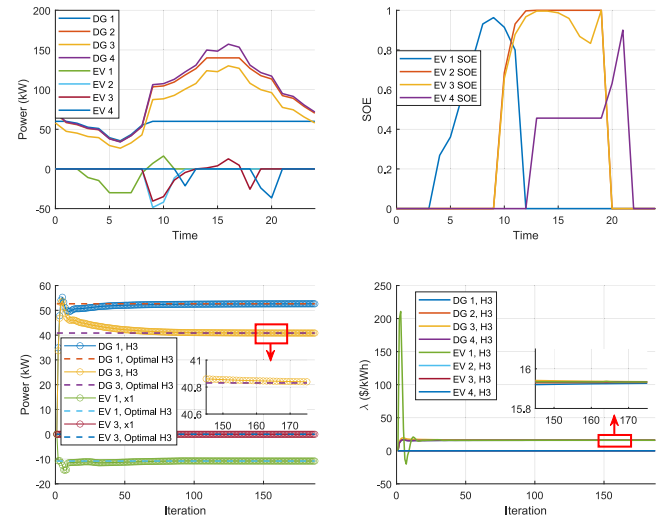

Fig. 5. Performance of the proposed optimal distributed economic dispatch algorithm in a time-varying network with compensation. (a) Optimal power generation schedules over 24 h for four DGs and four EVs. (b) State of energy profiles of EVs. (c) Charging and discharging power at hour 3. (d) Convergence of incremental cost, $\lambda_i^3(k)$, across agents.

targeted schedules for different types of vehicles, prioritizing those with higher charging efficiency, lower charging cost functions, and longer pre-scheduled charging durations for V2G participation. Fig. 5(b) shows that the algorithm satisfies the charging requirements of electric vehicles. For instance, both EV2 and EV3 achieve their charging targets at hour 19, where the state of energy, SOE_i^{19}, equals 1. The reduced slope of the dashed charging curve when the SOE exceeds 0.85 indicates compliance with CC-CV charging constraints, which reduces the risk of exceeding the rated voltage and thereby extends battery life. Fig. 5(c) depicts the charging and discharging power at hour 3. The proposed algorithm achieves optimal charging and discharging schedules. For example, in the zoomed-in section, DG3 reaches its optimal discharging power of 40.83 kW. The dashed line represents the optimal schedule derived from centralized optimization, while the circle markers represent the schedule obtained from the proposed algorithm, demonstrating consistency and validating the algorithm's ability to achieve optimal charging and discharging power. Finally, Fig. 5(d) shows the convergence of the incremental cost, $\lambda_i^3(k)$, across all agents, aligning at the optimal value of 15.9 $/kWh. This confirms the effectiveness of the proposed algorithm in achieving optimal cost distribution and further demonstrates the balance between supply and demand.

4.2. Time-varying network without compensation

In this case, similar to Case A, the same time-varying network is considered, but no compensation is applied using the proposed optimal distributed economic dispatch algorithm. As shown in Fig. 6(a) and (b), optimization results similar to Case A can be obtained without compensation. However, Fig. 6(c) and (d) illustrate that due to the impact of the time-varying network, the power output and incremental cost values oscillate continuously during the optimization process. The algorithm fails to converge and stops only after reaching the preset maximum iteration limit of 500. Compared to the 187 iterations required in Case A, an additional 313 steps are performed, leading to a significant waste of computational resources. More critically, the lack of convergence indicates that the optimization algorithm is not usable, as it cannot guarantee that the obtained solution is optimal.

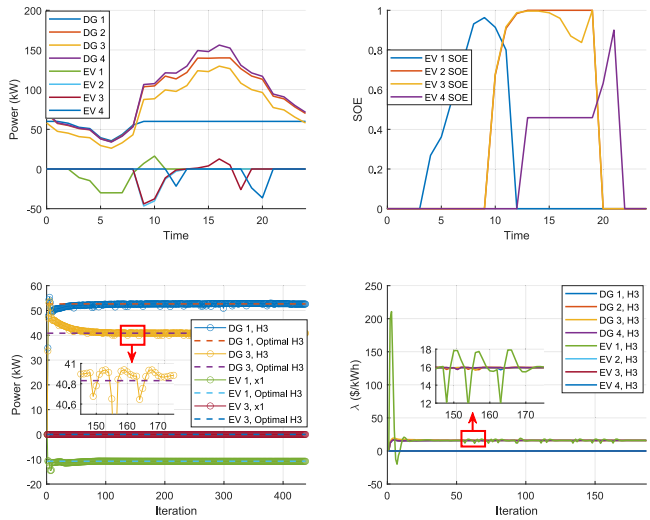

Fig. 6. Performance of the proposed optimal distributed economic dispatch algorithm in a time-varying network without compensation. (a) Optimal power generation schedules over 24 h for four DGs and four EVs. (b) State of energy profiles of EVs. (c) Charging and discharging power at hour 3. (d) Convergence of incremental cost, $\lambda_i^3(k)$, across agents.

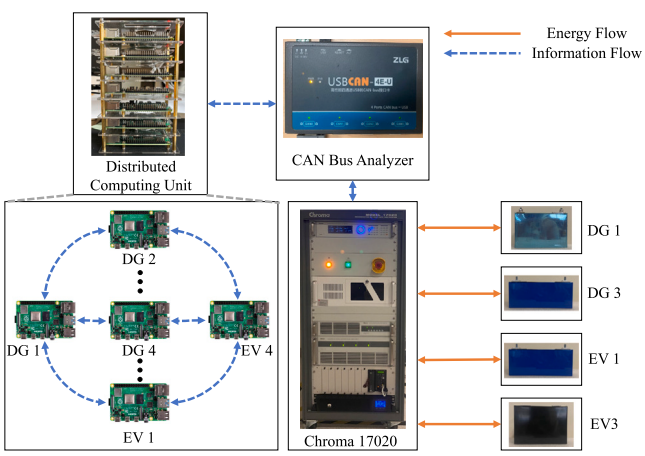

Fig. 7. Experimental platform of distributed economic dispatch.

5. Hardware experiment

The distributed economic dispatch system in the experiment consists of a distributed optimization computing network, a charge/discharge device connected to the real power grid, a CAN signal analyzer, and four chargeable electric vehicle battery cells, as shown in Fig. 7. The distributed optimization network is composed of 8 Raspberry Pi Model B devices (2 GB LPDDR4 RAM and 64-bit quad-core Cortex-A72 processors), as shown in Fig. 7.

Each Raspberry Pi is equipped with Python and the academic version of Gurobi. Python is responsible for calling the toolkits to build the distributed communication network, while Gurobi is used for optimization calculations. The computed optimal values are converted into CAN signals via the CAN Bus analyzer and sent to the Chroma 17020 charge/discharge device to regulate battery operations. Two fully charged batteries simulate generators, while two additional batteries with preset initial states represent electric vehicles. This setup emulates real-world DG and EV units, enabling distributed coordination, optimization, and economic dispatch.

The experiments are designed to validate the feasibility and effectiveness of the proposed distributed economic dispatch algorithm. In this system, Raspberry Pi devices act as independent computational units, performing the role of agents. These agents exchange λ values with neighboring units to achieve optimal economic dispatch, with different optimization calculations distinguished by timestamps. Once the algorithm converges, the Raspberry Pi, acting as the master controller, transmits the iteration results as control signals to a CAN bus analyzer. The control signals are then converted into CAN signal patterns via the CAN bus and sent to the Chroma 17020 device, which controls the charging and discharging of the electric vehicle batteries, completing the distributed optimization economic dispatch. This setup intentionally separates the computational layer, which consists of the 8 Raspberry Pi agents, from the physical execution layer, which uses the 4 Chroma channels, to validate the distributed optimization algorithm itself. The mapping is as follows: all 8 computation nodes, the Raspberry Pi units, participate in the distributed optimization, exchanging data to find a converged solution. Once converged, the final dispatch commands are sent via the CAN bus to the physical devices. It is worth noting that the Chroma 17020 charging/discharging device supports only four battery channels. Therefore, despite having 8 Raspberry Pi devices performing optimization calculations in the distributed network, only signals from 4 of these Raspberry Pis are transmitted to the Chroma 17020. This hardware limitation does not impact the validation of the optimization algorithm. The process of sending the final control signal from any number of computation nodes to the physical device and its subsequent execution is a microsecond-level operation, which does not constitute a bottleneck for the algorithm's convergence time. The experiment's primary focus is the distributed computation, which is successfully performed by all 8 agents. In Fig. 8(a), the comparison between actual and simulated power in the distributed economic dispatch illustrates their strong agreement. Fig. 8(b) highlights that in the distributed system implemented using Raspberry Pi, the proposed algorithm stabilizes the incremental cost λ at the optimal value of 15.9 $/kWh at hour 3. Since EV3 has not yet arrived by this time, it is excluded from the rolling optimization, resulting in a λ value of zero for EV3. These experimental findings confirm the feasibility and effectiveness of the proposed method.

Each Raspberry Pi is configured with Python and the Gurobi solver, utilizing the ZeroMQ (ZMQ) library for inter-agent communication. A key advantage of the distributed approach is that each RPi only needs to solve a simple, single-agent optimization subproblem. The total solution time per iteration is composed of local computation time and communication time. Because the single-agent subproblem is a simple convex quadratic program, the computation time is in the millisecond range for DGs. The V2G subproblem is slightly more complex due to the bidirectional charging and discharging modes, but its solution time remains low, on the order of tenths of a second. The proposed algorithm uses synchronous communication, requiring agents to confirm consistent timestamps with neighbors before proceeding. As the number of neighbors increases, this synchronization overhead also grows but is similarly on the order of tenths of a second per iteration. In this experiment, at $t = 0$, with only 4 DG agents participating, the algorithm converged in 117 iterations, taking 3.58 s. At $t = 4$, the problem size increased to 7 agents (4 DGs and 3 EVs), requiring 143 iterations and 17.43 s. At $t = 8$, with all 8 agents active, the system converged in 71 iterations, taking 21 s. In summary, while the distributed approach effectively decomposes the large-scale problem, the synchronous communication required for consensus can increase the total solution time. Therefore, developing asynchronous communication mechanisms based on this framework is a critical direction for future research.

6. Conclusion

This study presents a novel approach to addressing the challenges of distributed economic dispatch in V2G systems. The proposed method transforms the MIQP problem into a convex quadratic programming

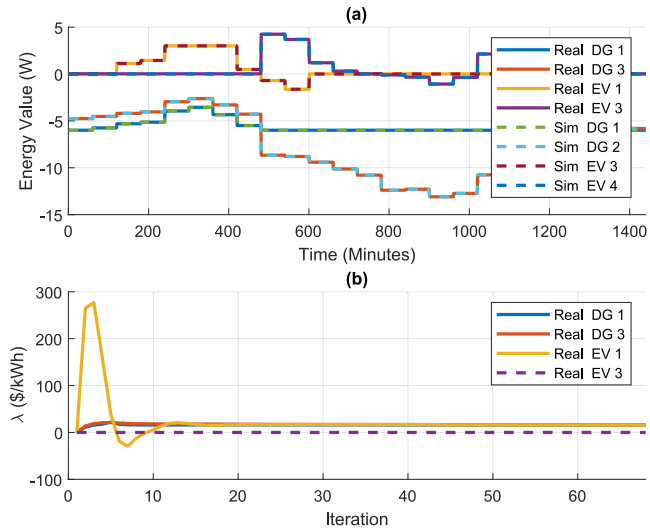

Fig. 8. Experimental result of proposed optimal distributed economic dispatch algorithm in the time-varying network with compensation. (a) Comparison of real and simulated power over 24 h. (b) Convergence of incremental cost iteration of the real world.

problem and then combines ADMM with MPC to enable rolling optimization. This facilitates the plug-and-play integration of V2G systems. By extending Lagrange multipliers to vector form, the approach effectively manages time-state coupling constraints, which are often overlooked in existing methods. Furthermore, the incorporation of physical constraints, such as CC-CV charging mode and charge–discharge efficiency factors, ensures the longevity of V2G electric vehicle batteries, addressing key concerns in real-world applications. A dispatch packet strategy is employed to alleviate communication and computational overheads, making the proposed algorithm both scalable and efficient for large-scale deployment. The feasibility and effectiveness of the approach are validated through both simulation and physical experiments, demonstrating its potential to optimize V2G coordination in dynamic, distributed environments. Future research will explore both the application of machine learning techniques to predict dynamic network conditions and the extension of this algorithm to more complex integrated energy systems. These future models will aim to incorporate diverse resources, such as hydrogen and hydro power, while also addressing significant challenges like renewable energy uncertainty and additional physical constraints, including frequency stability. It is also important to acknowledge the assumptions of the current model as limitations. The proposed MPC framework, for example, relies on the quality of short-term predictions, and the economic dispatch model does not account for physical line losses, which are important considerations for practical deployment in larger networks. Regarding practical limitations, a key observation from the hardware experiments is that the algorithm's synchronous nature is sensitive to computational heterogeneity; slower agents, such as aging devices, can bottleneck the entire system's convergence time. Therefore, a critical direction for future work is to develop asynchronous optimization mechanisms that mitigate the impact of such stragglers while maintaining rapid convergence.

CRediT authorship contribution statement

Chuan-Fan Lu: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Guo-Ping Liu:** Supervision, Resources, Funding acquisition. **Rui Yan:** Software, Data curation. **Jinqiang Cui:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Proof of Remark 1

The proof proceeds by contradiction. It will be demonstrated that for the cost-minimization problem (14), any solution that includes simultaneous charging and discharging ($P_i^{t,dis} > 0$ and $P_i^{t,cha} > 0$) for an EV $i \in \mathcal{M}$ cannot be optimal if the efficiencies $\eta_i^{dis} < 1$ and $\eta_i^{cha} < 1$ [22].

Assume $\{P^\star\}$ is an optimal solution to problem (14) (formulated without the constraint (15)). Suppose this solution, for some EV $i \in \mathcal{M}$ at time $t \in \mathcal{T}$, violates the condition, such that $P_i^{t,dis,\star} > 0$ and $P_i^{t,cha,\star} > 0$. By the problem's premise, the efficiency product $\eta_i^{dis}\eta_i^{cha} < 1$.

A new candidate solution $\{\hat{P}\}$ can be constructed. This solution is identical to $\{P^\star\}$ in all variables, except for those of EV i and at least one generator $j \in \mathcal{N}$ at time t. Let $\epsilon > 0$ be a sufficiently small positive constant satisfying both $\epsilon \le P_i^{t,cha,\star}$ and $\epsilon\eta_i^{cha}\eta_i^{dis} \le P_i^{t,dis,\star}$. Such an ϵ must exist, as both $P_i^{t,cha,\star}$ and $P_i^{t,dis,\star}$ are strictly positive by hypothesis, and $\eta_i^{cha}\eta_i^{dis} > 0$. The new power levels for EV i at time t are defined as:

$$\hat{P}_i^{t,cha} = P_i^{t,cha,\star} - \epsilon, \tag{A.1}$$

$$\hat{P}_i^{t,dis} = P_i^{t,dis,\star} - \epsilon\eta_i^{cha}\eta_i^{dis}. \tag{A.2}$$

First, the feasibility of $\{\hat{P}\}$ is verified. The internal battery power rate $\hat{P}_i^{t,\text{bat}}$ for EV i is calculated using Eq. (10):

$$\begin{aligned}\hat{P}_i^{t,\text{bat}} &= \frac{1}{\eta_i^{dis}}\hat{P}_i^{t,dis} - \eta_i^{cha}\hat{P}_i^{t,cha}\\ &= \frac{1}{\eta_i^{dis}}(P_i^{t,dis,\star} - \epsilon\eta_i^{cha}\eta_i^{dis}) - \eta_i^{cha}(P_i^{t,cha,\star} - \epsilon)\\ &= \left(\frac{1}{\eta_i^{dis}}P_i^{t,dis,\star} - \eta_i^{cha}P_i^{t,cha,\star}\right) - \epsilon\eta_i^{cha} + \epsilon\eta_i^{cha}\\ &= P_i^{t,\text{bat},\star}.\end{aligned} \tag{A.3}$$

Since $\hat{P}_i^{t,\text{bat}} = P_i^{t,\text{bat},\star}$, the state of energy SOE_i^t trajectory determined by (11) is unaffected. Consequently, all inter-temporal constraints for EV i remain satisfied.

Next, the impact on the power balance constraint (3) is examined. The net power supplied by EV i is $P_i^t = P_i^{t,dis} - P_i^{t,cha}$. The change in this net power is:

$$\begin{aligned}\Delta P_i^t &= \hat{P}_i^t - P_i^{t,\star} = (\hat{P}_i^{t,dis} - \hat{P}_i^{t,cha}) - (P_i^{t,dis,\star} - P_i^{t,cha,\star})\\ &= (P_i^{t,dis,\star} - \epsilon\eta_i^{cha}\eta_i^{dis} - (P_i^{t,cha,\star} - \epsilon))\\ &\quad - (P_i^{t,dis,\star} - P_i^{t,cha,\star})\\ &= \epsilon(1 - \eta_i^{cha}\eta_i^{dis}).\end{aligned} \tag{A.4}$$

Given that $\epsilon > 0$ and $\eta_i^{cha}\eta_i^{dis} < 1$, the net power change ΔP_i^t is strictly positive. This represents a net power surplus or increased supply gained by reducing the inefficient simultaneous charge/discharge cycle.

To maintain the system-wide power balance (3), this surplus ΔP_i^t must be offset. This can be achieved by reducing the power output of a generator $j \in \mathcal{N}$:

$$\hat{P}_j^t = P_j^{t,\star} - \Delta P_i^t = P_j^{t,\star} - \epsilon(1 - \eta_i^{cha}\eta_i^{dis}). \tag{A.5}$$

This adjustment is assumed to be feasible, as (6) generally ensures the system is not operating at its absolute minimum generation floor.

Finally, the total cost is assessed. The change in the total system cost (14) arises from the adjustment to generator j's output. As defined in (2), the generator's cost function $C_j(\cdot)$ is strictly convex (since $a_j > 0$).

Because $\Delta P_i^t > 0$, the new generator output $\hat{P}_j^t$ is strictly less than $P_j^{t,\star}$. By the property of strict convexity for C_j:

$$C_j(\hat{P}_j^t) < C_j(P_j^{t,\star}). \tag{A.6}$$

This implies that the total production cost of the new feasible solution $\{\hat{P}\}$ is strictly lower than the cost of $\{P^\star\}$.

This result contradicts the initial hypothesis that $\{P^\star\}$ is an optimal solution. Therefore, any optimal solution must inherently satisfy $P_i^{t,dis} \cdot P_i^{t,cha} = 0$ for all $i \in \mathcal{M}, t \in \mathcal{T}$. The proof is complete.

Data availability

No data was used for the research described in the article.

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